



Battery container simulation on Ansys

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Contents

1	Introduction	2
2	Rules and Regulations from formula student 2025 rules	2
2.1	Segment and Mass Constraints	2
2.2	Material Baseline Thicknesses	2
2.3	Structural Geometry Rules	3
2.4	Simulation Environment and Fasteners	3
3	Assumptions, Methodology	3
4	Implementation	4
4.1	Geometry	4
4.2	Material Characteristics	4
4.3	Connexion	5
4.3.1	Contact	5
4.4	Loading and Boundary Conditions	5
4.4.1	Gravity and Mass Representation	5
4.4.2	Fixed Supports	5
4.4.3	Acceleration Loads	6
4.5	Meshing Strategy	6
5	Results	7
5.1	Total Deformation	7
5.2	Directional Deformation	8
5.3	Equivalent (Von Mises) Stress	8
6	Discussion and Analysis	9
7	Conclusion	9
8	Future Work	9

1 Introduction

This section presents the finite element analysis (FEA) of a battery container performed using Ansys Mechanical. The objective is to evaluate the structural response of the container under realistic loading conditions, including gravity and acceleration.

2 Rules and Regulations from formula student 2025 rules

The TSAC structure, its mounting to the chassis, and the internal mounting of every cell must withstand the following inertial accelerations simultaneously or in independent load cases as specified in the ASES

Rule : T9.2.1:

- **Longitudinal Direction (Forward/Aft):** 40 *g*
- **Lateral Direction (Left/Right):** 40 *g*
- **Vertical Direction (Up/Down):** 20 *g*

2.1 Segment and Mass Constraints

Each individual accumulator segment within the container must respect the following limits :

Rule : EV5.3

- **Maximum Mass:** 12 kg
- **Maximum Static Voltage:** 120 VDC
- **Maximum Energy:** 6 MJ

2.2 Material Baseline Thicknesses

If metallic materials are used, the following minimum thicknesses must be respected to avoid further equivalency proofing. Any thickness below these requires simulation/testing data in the ASES :

Component	Steel (Min)	Aluminium (Min)
Container Bottom	1.25 mm	3.2 mm
Vertical Walls	0.9 mm	2.3 mm
Lid and Covers	0.9 mm	2.3 mm

2.3 Structural Geometry Rules

- **Internal Walls:** Segments must be separated by internal vertical walls that extend upwards until the lid .
- **Cutouts/Holes:** The total cutout area (for cooling or wiring) must be **less than 25%** of the area of the respective single wall.
- **Pouch Cell Fixation:** Pouch cells must be fixed on at least **80%** of their large surfaces.

2.4 Simulation Environment and Fasteners

- **Ambient Temperature:** For non-metallic materials (excluding continuous fiber laminates), calculations must assume 60 °C .
- **Attachment Points:** A minimum of **2 attachment points** to the chassis is required .
- **Mounting Brackets:** Must be steel (≥ 1.6 mm) or aluminium (≥ 4 mm) with gussets for bending loads .
- **Edge Distance (e/D):** All bolted joints must respect an edge distance ratio of 1.5 or greater .

3 Assumptions, Methodology

- The battery modules are modeled as simplified rectangular volumes.
- The cells inside each module are not modeled individually.
- Each module is assigned an equivalent material with a density computed as:

$$\rho = \frac{m_{\text{cells}}}{V_{\text{module}}}$$

- The weight of the cells is represented using distributed mass and gravity.

- Contact between modules and the container is simplified (bonded assumption).

4 Implementation

4.1 Geometry

- The geometry is designed in Onshape and imported into Ansys.
- Container dimensions: $590 \times 670 \times 270$ mm
- Module dimensions: $190 \times 371 \times 84.7$ mm
- Spacing between modules: 32.741 mm
- All components are modeled as rectangular prisms.

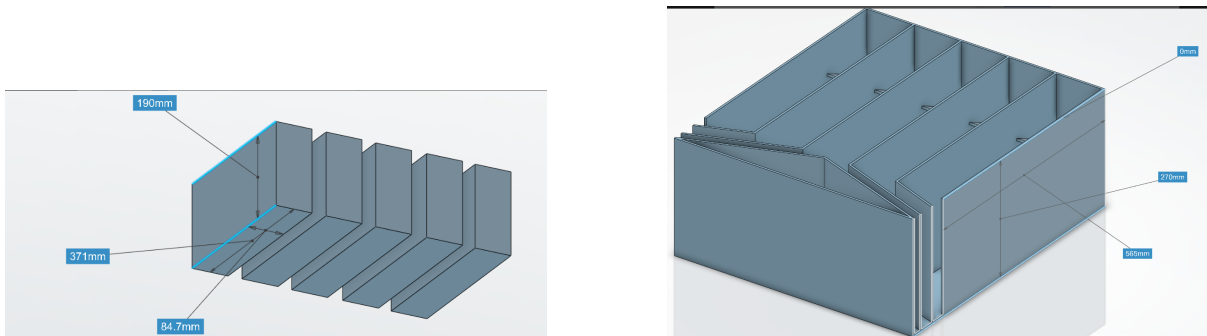


Figure 1: Shapes

4.2 Material Characteristics

The container is made of aluminum to ensure realistic stiffness and strength behavior.

The modules are assigned an equivalent material based on their density. A material close to this density is selected. In this case, Balsa wood is used with:

$$\rho = 254 \text{ kg/m}^3$$

4.3 Connexion

4.3.1 Contact

The contact is modeled between each module and its corresponding surface on the battery container. The type of contact is **Bonded** because:

- The module is directly screwed to the container.
- There is no gap between the module and the bottom surface.
- There is no sliding between the surfaces.
- The load is transferred through contact pressure.

4.4 Loading and Boundary Conditions

4.4.1 Gravity and Mass Representation

The mass of the cells is represented using a distributed mass applied to the module regions.

Gravity is applied as follows:

- Direction: $-Z$
- Magnitude: 9.81 m/s^2

This approach ensures that the weight of the cells is naturally transferred to the bottom surface of the container.

4.4.2 Fixed Supports

Fixed supports are applied to represent the attachment of the container to the vehicle structure:

- Along the upper mounting edges

- On selected bottom regions

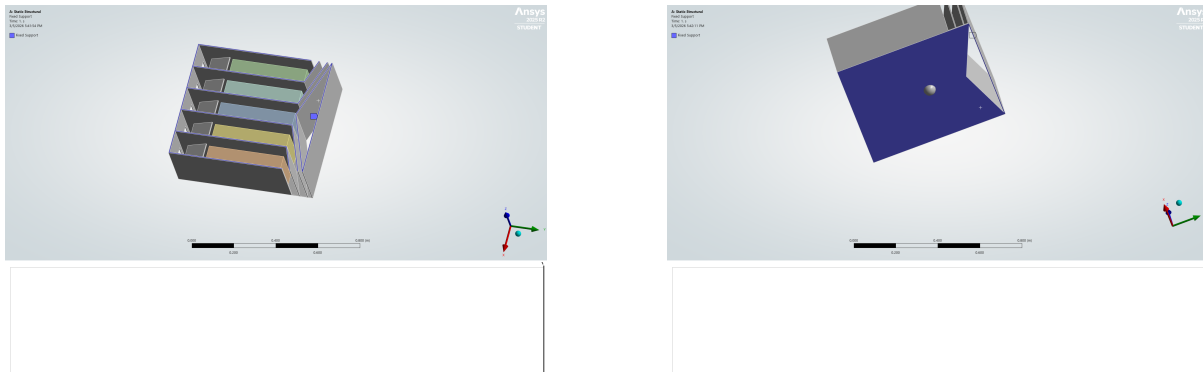


Figure 2: Boundary conditions (fixed supports)

4.4.3 Acceleration Loads

Acceleration loads are applied to simulate extreme operating conditions:

- $40g$ in the longitudinal direction (x)
- $40g$ in the lateral direction (y)
- $20g$ in the vertical direction (z)

4.5 Meshing Strategy

First simulation is done with the default meshing. A refined mesh is applied in critical areas, especially on the bottom surface where stress concentrations are expected.

- Element size on bottom surface: 0.01 m



Figure 3: Mesh configuration

5 Results

5.1 Total Deformation

The total deformation highlights the global stiffness of the structure, with maximum deformation occurring in unsupported regions.

The maximum deformation is 9.23×10^{-6} m ($9.23 \mu\text{m}$), which is acceptable for this application. For acceleration along the x-direction, deformation occurs primarily on the lateral side. In reality, this lateral side will be maintained fixed by connection to another component, so this aspect requires further detailed study.

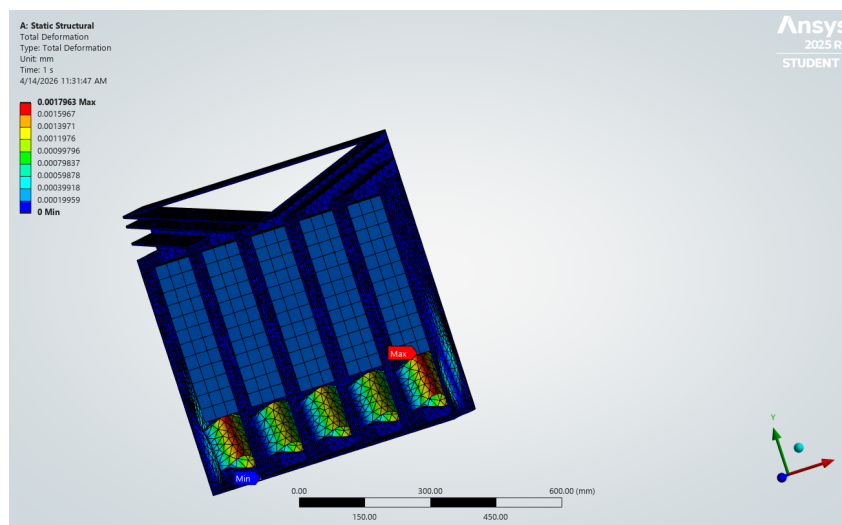


Figure 4: Total deformation

5.2 Directional Deformation

This is a useful plot because it shows the deformation along the X axis, not just the total displacement. The values are very small, from about -0.000503 mm to +0.000503 mm, so the part is only moving slightly in the global X direction

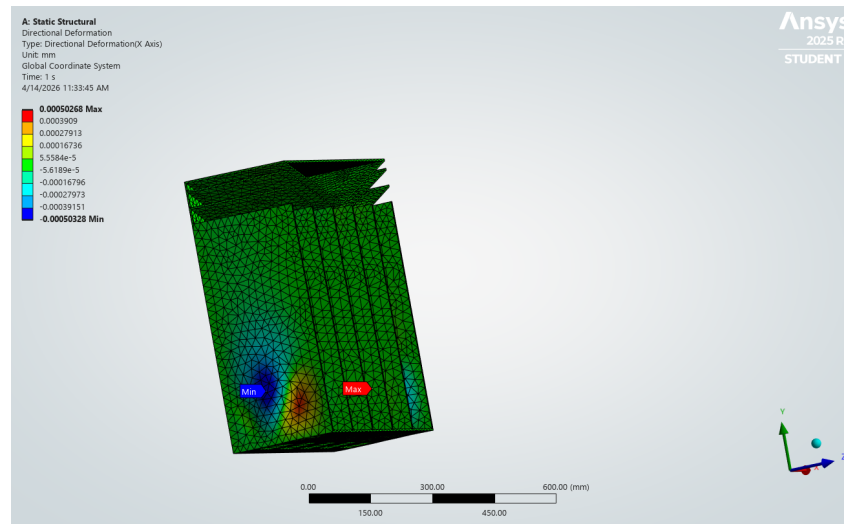


Figure 5: Directional deformation

5.3 Equivalent (Von Mises) Stress

This equivalent von Mises stress result looks very low overall: the legend shows a maximum of about 0.36969 MPa and a minimum near zero, so the part is far from yielding. Because the peak stress is only around 0.37 MPa and the fact that the material is aluminum the model is in a very safe elastic range

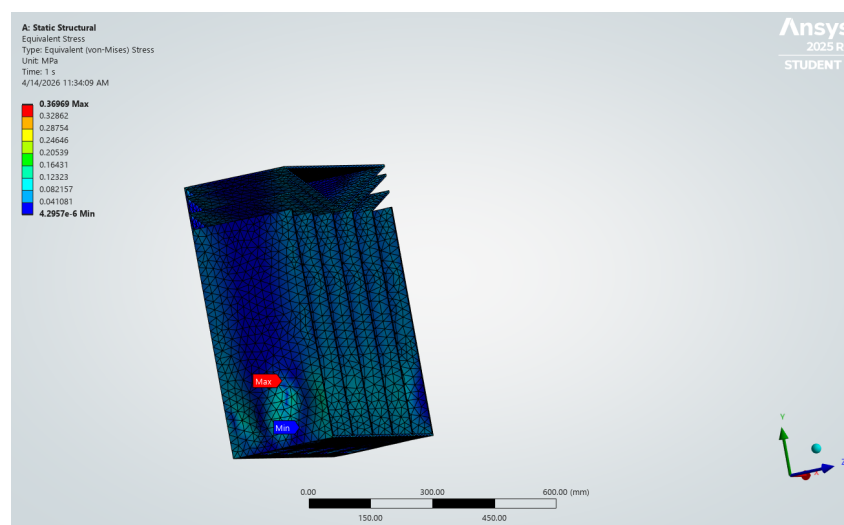


Figure 6: Equivalent (Von Mises) stress

6 Discussion and Analysis

Discuss the significance of the results. Consider:

- Are we obliged to simulate the bolts in the contact between the bottom surface and the module?
- Challenges encountered
- Possible improvements

7 Conclusion

The overall results are good since the deformations are very low. The results also fit well with the characteristics of aluminum. For a more accurate simulation, we will modify the surface of the module in contact with the bottom surface of the container to improve the transmission of forces between the two parts.

8 Future Work

Change the container design to simulate the actual contact between the module and the bottom surface. The part of the module that will be in contact with the container's bottom surface consists of two thin, wide surfaces, which can be modeled as two lines representing the contact between the module and the bottom surface.

Also, integrate the container into the vehicle model and run a realistic simulation with the car to observe the behavior and see what happens.